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Lab Assignment
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ENCS304 : Computer Networks

LAB Assignment 1

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Experiment	Basic Network Topologies Using Switches and Hubs

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ENCS304 : Computer Networks | LAB Assignment 1

Basic Network Topologies Using Switches and Hubs

Objectives

The primary objective of this experiment is to design and analyse three fundamental LAN topologies - Star, Bus-like (Hub), and Ring-like (Loop) - using Cisco Packet Tracer. Through hands-on simulation, the experiment develops a clear understanding of how physical network layout influences connectivity, performance, collision behaviour, and fault tolerance.

- Configure end devices with valid IPv4 addresses within the 192.168.10.0/24 subnet.
- Verify end-to-end connectivity across each topology using ICMP ping.
- Observe and compare packet propagation behaviour in switched versus hub-based environments.
- Analyse fault-tolerance by simulating a link failure in the ring-like topology.
- Draw practical conclusions regarding the suitability of each topology for real-world LAN deployment.

Task 1 : Star Topology (Switch)

Task 1: Star Topology using a 2960-24TT Switch

Network Diagram

Four PCs are connected to a central 2960-24TT switch using Copper Straight-Through cables. Each PC was assigned a static IPv4 address in the 192.168.10.0/24 subnet. The switch operates at Layer 2 and forwards frames only to the intended destination port, eliminating unnecessary traffic on other ports.

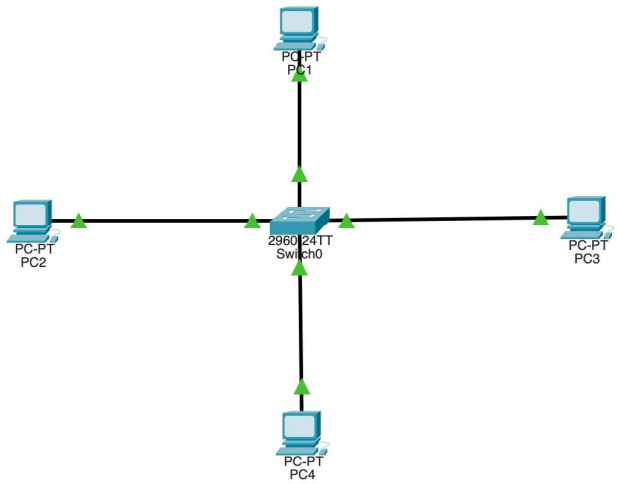


Figure 1: Star Topology - 4 PCs connected to Switch0 (all links active, green)

IP Address Configuration

Device	IP Address	Subnet Mask	
PC1	192.168.10.1	255.255.255.0	
PC2	192.168.10.2	255.255.255.0	
PC3	192.168.10.3	255.255.255.0	
PC4	192.168.10.4	255.255.255.0	

Ping Results

Pings were executed from PC1 to PC2, PC3, and PC4. All returned 4/4 successful replies with 0% packet loss. The higher latency on the first ping to PC3 (23ms) is due to ARP resolution occurring for the first time; subsequent packets dropped to under 1ms, confirming efficient unicast switching.

Ping Results (in Command Prompt)

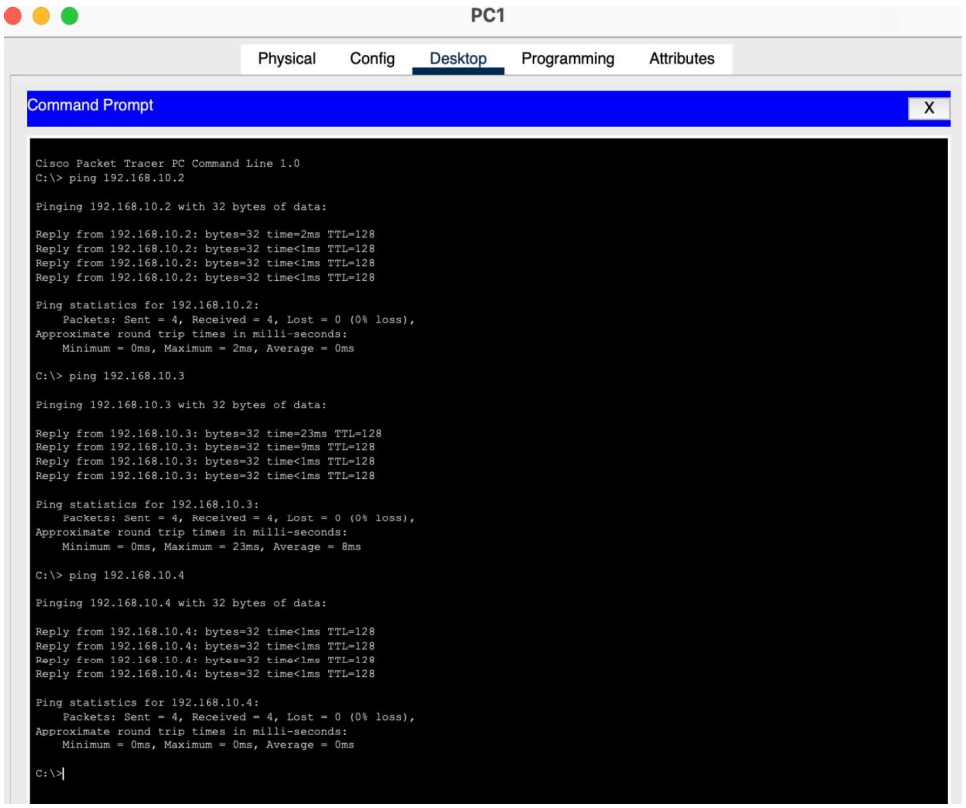


Figure 2: PC1 Command Prompt — Pings to PC2, PC3, PC4 (all 4/4 successful, 0% loss)

Task 2 : Bus-like Topology (Hub)

Task 2: Bus-like Topology using a Hub-PT

Network Diagram

The switch from Task 1 was replaced with a Hub-PT. The hub operates at Layer 1 (Physical Layer) and broadcasts every incoming frame to all connected ports, creating a single shared collision domain. IP addressing remained identical to Task 1.

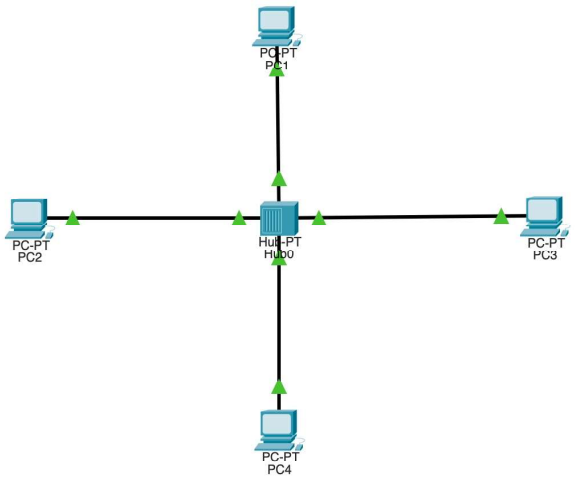


Figure 3: Bus-like Topology — 4 PCs connected to Hub0 (all links active, green)

Ping Results

Pings from PC1 to all other hosts succeeded with 0% loss. Unlike the switch, the hub floods every frame to all ports simultaneously. In Packet Tracer Simulation Mode, this broadcast behaviour is clearly visible - every PC receives every packet regardless of whether it is the intended destination, which would cause collisions under concurrent load.

Ping Results (in Command Prompt)

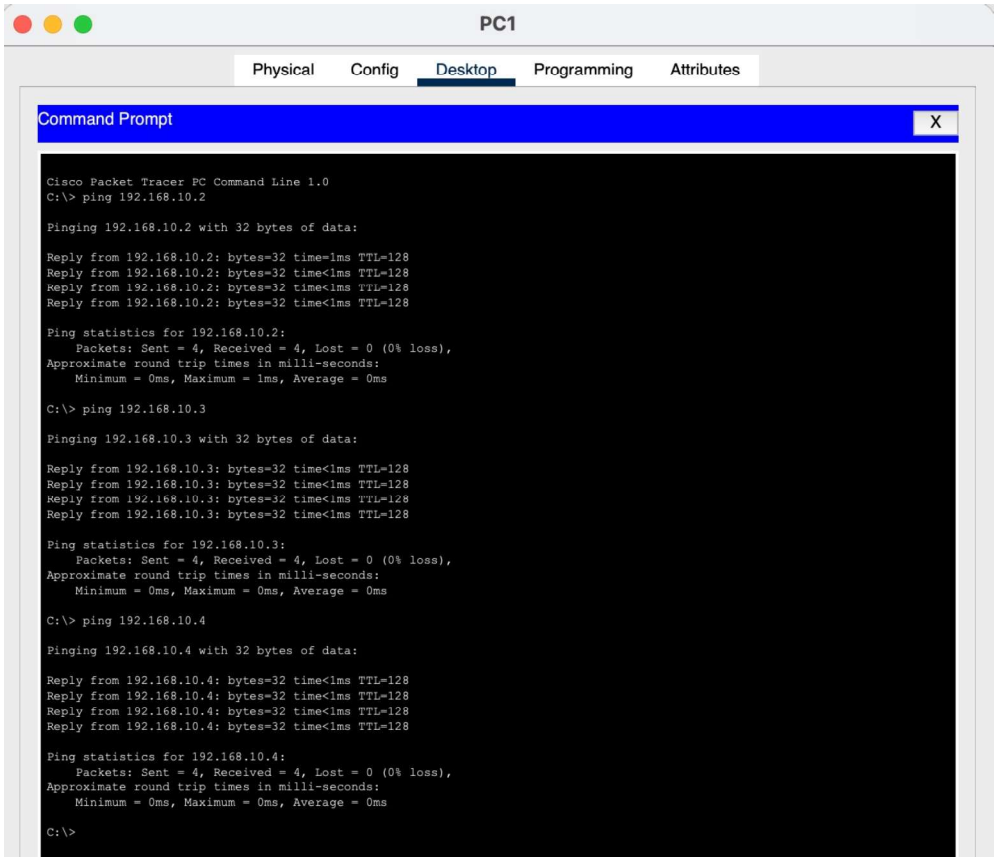


Figure 4: PC1 Command Prompt — Ping results via Hub (0% loss, 1ms latency)

Task 3 : Ring-like Topology (3 Switches in Loop)

Task 3: Ring-like Topology - 3 Switches connected in a triangle loop

Network Diagram

Three 2960-24TT switches (SwitchA, SwitchB, SwitchC) were interconnected in a triangular loop. PC1 and PC2 were connected to SwitchA, PC3 and PC4 to SwitchB, and PC5 to SwitchC. The orange dot on SwitchB is Spanning Tree Protocol (STP) intentionally blocking one port to prevent broadcast storms - this is correct and expected behaviour in a loop topology.

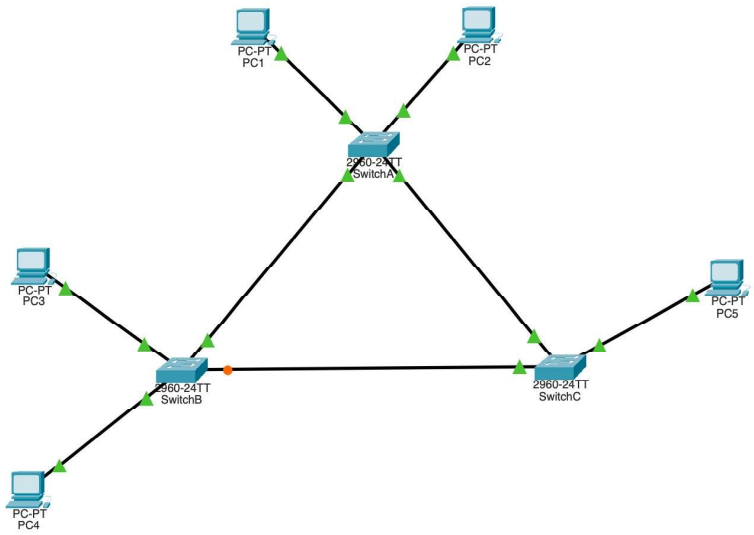


Figure 5: Ring-like Topology - 3 switches in triangle loop with 5 PCs. Orange dot = STP blocked port (normal).

IP Address Configuration

Device	IP Address	Subnet Mask	Connected Switch
PC1	192.168.10.1	255.255.255.0	SwitchA
PC2	192.168.10.2	255.255.255.0	SwitchA
PC3	192.168.10.3	255.255.255.0	SwitchB
PC4	192.168.10.4	255.255.255.0	SwitchB
PC5	192.168.10.5	255.255.255.0	SwitchC

Ping Results

Pings from PC1 to PC3, PC4 (on SwitchB) and PC5 (on SwitchC) all returned 4/4 replies with 0% loss, confirming full cross-switch connectivity. The slightly higher first-packet latency to PC3 (18ms) reflects inter-switch ARP resolution.

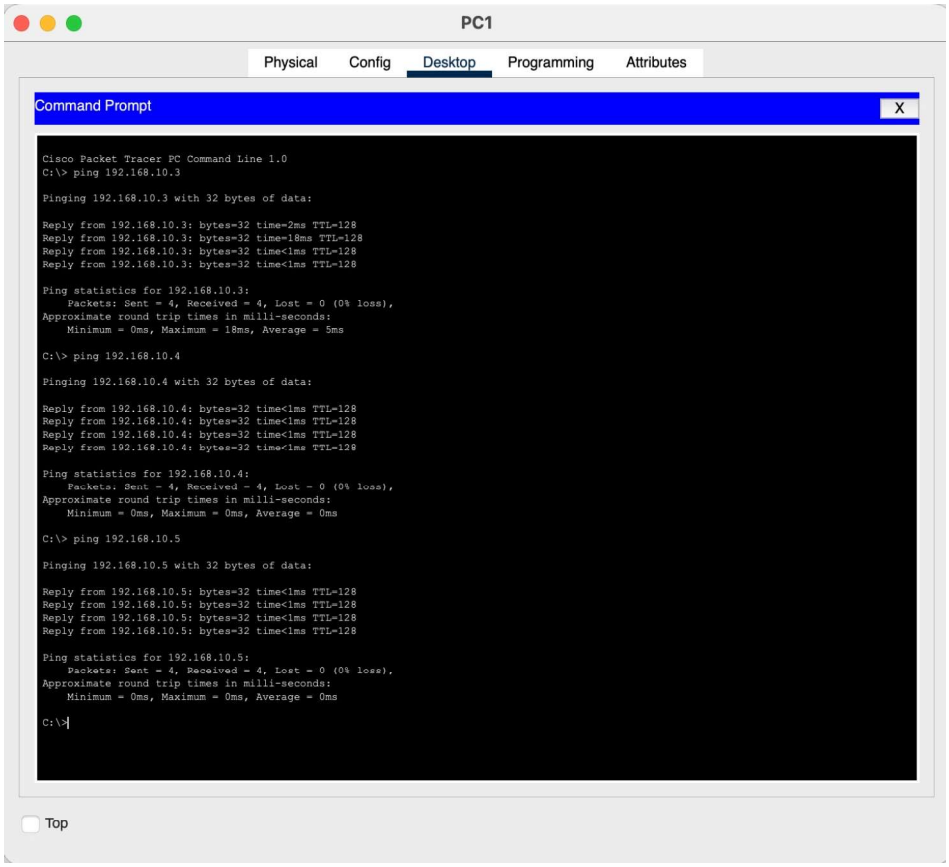


Figure 6: PC1 Command Prompt — Cross-switch pings to PC3, PC4, PC5 (all 4/4 successful, 0% loss)

Task 4 : Failure Test

Task 4: Link Failure Test - Cable removed between SwitchA and SwitchB

Test Setup

To simulate a physical link failure, the cable between SwitchA and SwitchB was deleted. This forces all traffic between PC1/PC2 (SwitchA) and PC3/PC4 (SwitchB) to reroute through SwitchC. The topology below shows the two remaining inter-switch links after the deletion.

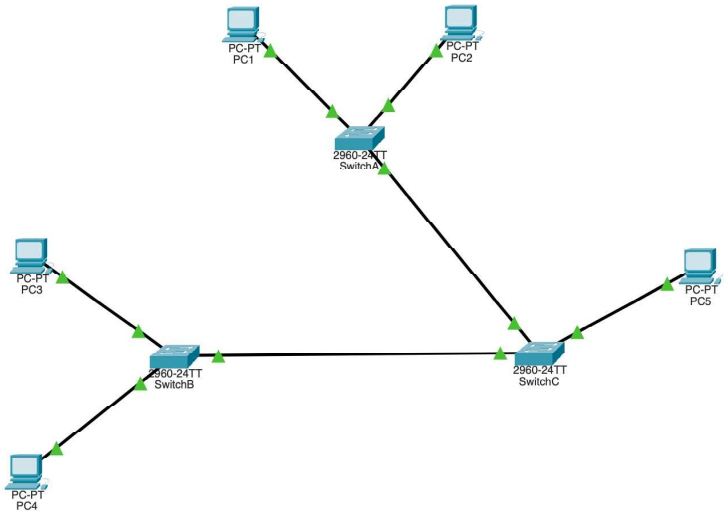
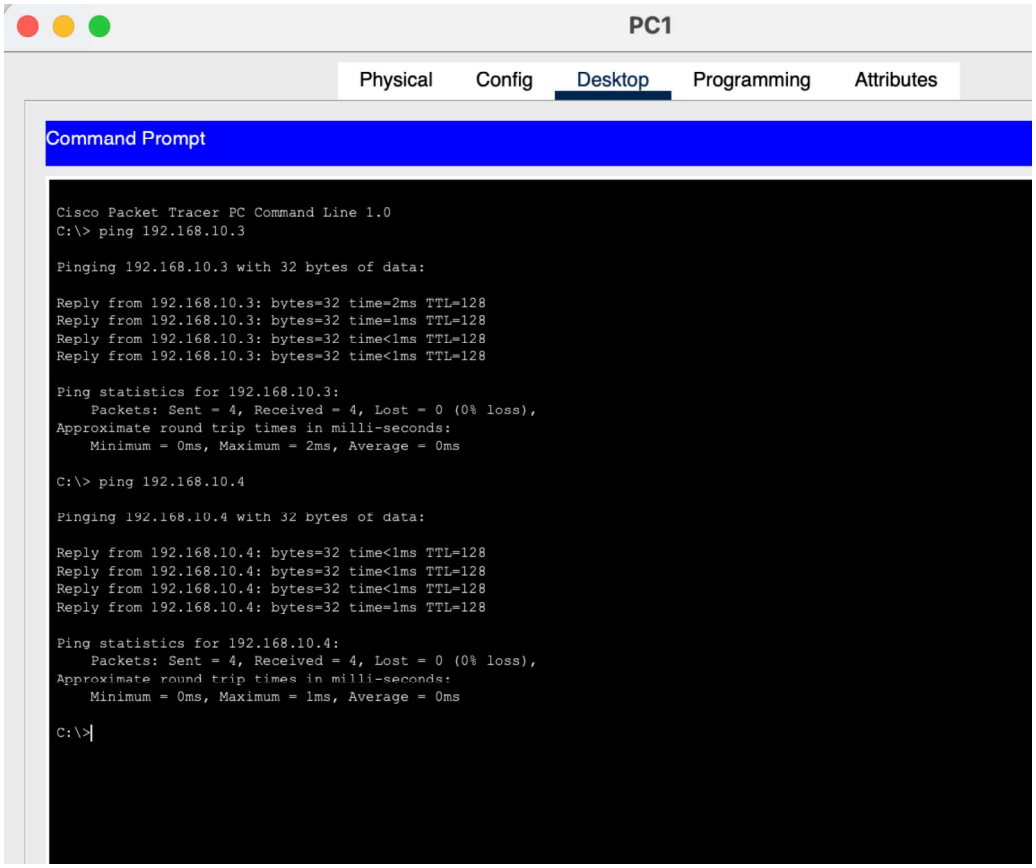


Figure 7: Failure Test Topology - SwitchA–SwitchB link removed. Only 2 inter-switch links remain.

Ping Results After Link Failure

Pings from PC1 to PC3 and PC4 were re-issued after the link removal. All returned 4/4 successful replies with 0% loss. No timeouts were observed because STP had already pre-converged - when the SwitchA–SwitchB link was removed, STP instantly unblocked the previously blocked SwitchB–SwitchC port, establishing the alternate path before the pings were issued. This demonstrates excellent fault tolerance of the ring-like topology.

Ping Results (in Command Prompt)



6. Comparative Observations

Aspect	Star (Switch)	Bus-like (Hub)	Ring-like (Loop)
Traffic Handling	Unicast to target port only	Broadcasts to all ports	Unicast via shortest path
Collision Domain	Isolated per port - no collisions	Single shared domain	Isolated per switch segment
Ping Latency	< 1ms consistently	1ms (sequential)	< 1ms including after failover
Fault Tolerance	None - single point of failure	None	Yes - STP reroutes instantly
Packet Loss (observed)	0%	0%	0% (including after link failure)
Real-world Use	Office & campus LANs	Legacy / low-cost setups	Data centres, backbone links

7. Conclusions

- This experiment successfully demonstrated the construction, IP configuration, and analysis of three fundamental LAN topologies in Cisco Packet Tracer. The following conclusions were drawn from the actual test results:

- Star topology with a switch delivered the best general-purpose performance. All pings completed with 0% loss and sub-millisecond latency. The switch's MAC address table ensures unicast forwarding, eliminating unnecessary traffic on all other ports.

- Hub-based topology functioned correctly for sequential pings but is fundamentally limited by its shared collision domain. All ports receive every frame regardless of destination, making it unsuitable for high-traffic environments. It is considered obsolete in modern networking.

- Ring-like topology with 3 switches demonstrated superior fault tolerance. When the SwitchA–SwitchB link was removed, STP instantly reconverged and rerouted traffic through SwitchC with zero packet loss. The STP-blocked port (orange dot on SwitchB) is a deliberate feature that prevents broadcast storms while keeping the redundant path on standby for instant failover.

- In conclusion, star topology is recommended for standard office LANs, while ring or mesh topologies are preferable wherever high availability and fault tolerance are required, such as in data centres or critical infrastructure networks.